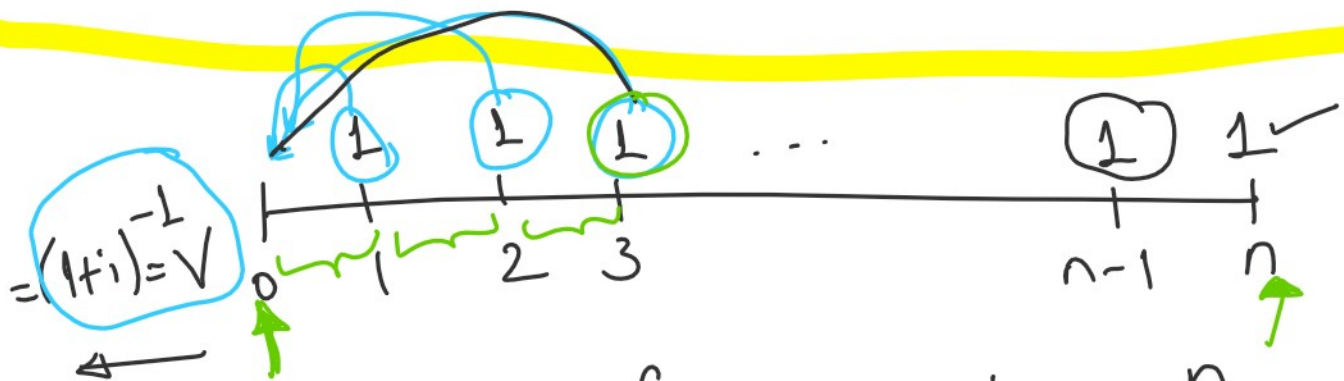


Annuities.



Number of payments = n

$$PV_{\text{annuity}} = 1 \cdot V + 1 \cdot V^2 + 1 \cdot V^3 + \dots + 1 \cdot V^n$$

V: common ratio.

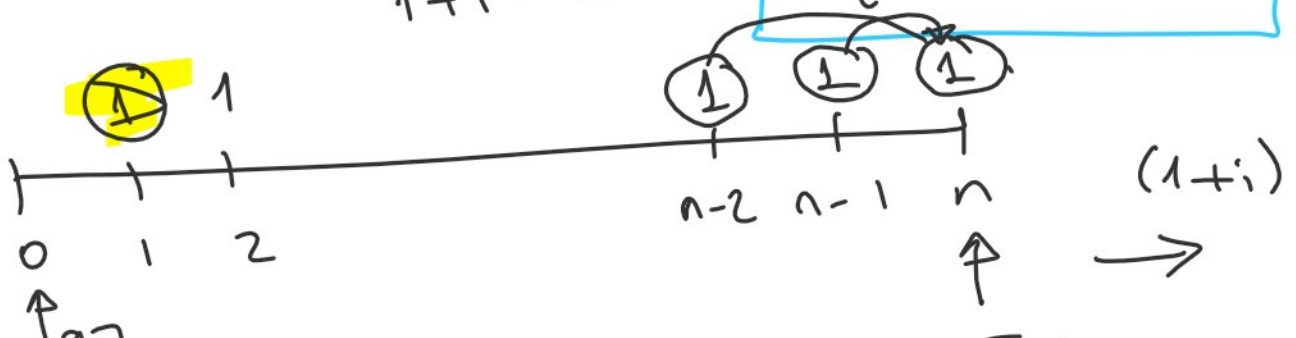
$$= V (1 + V + V^2 + \dots + V^{n-1})$$

$\frac{1-r^n}{1-r} = \text{G.S.S.}$

$$= V \cdot \frac{1 - V^n}{1 - V}$$

$$= \frac{1 - V^n}{\frac{1}{V}(1 - V)} = \frac{1 - V^n}{(1+i)(1 - \frac{1}{1+i})}$$

$$= \frac{1 - V^n}{1 + i - 1} = \boxed{\frac{1 - V^n}{i} = a_{\overline{n}|i}}$$



0 1 2
↑
 a_n

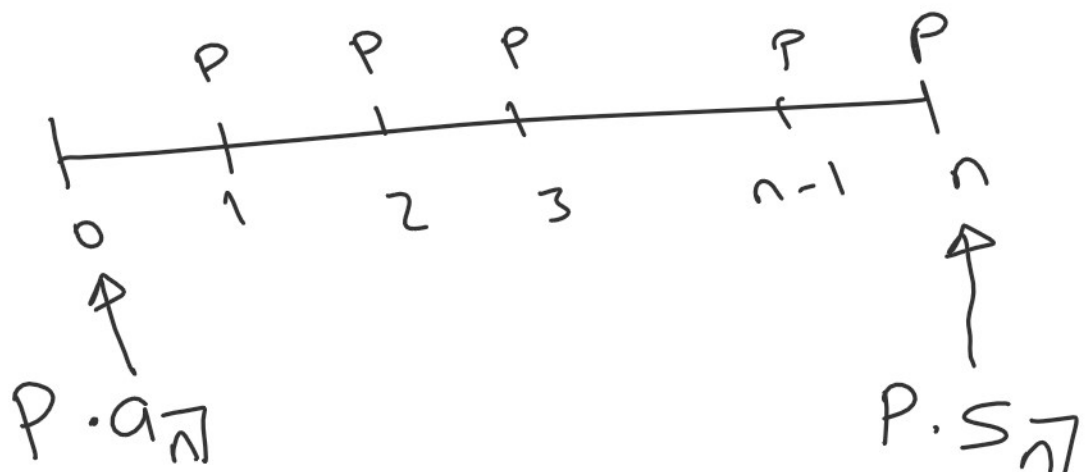
$\uparrow \rightarrow$
 S_n

common ratio = $1+i$

$$\begin{aligned} S_n &= 1 + 1 \cdot (1+i) + 1 \cdot (1+i)^2 + \dots + 1 \cdot (1+i)^{n-1} \\ &= \frac{1 - (1+i)^n}{1 - (1+i)} \\ &= \frac{1 - (1+i)^n}{1 - 1 - i} \\ &= \frac{1 - (1+i)^n}{-i} \end{aligned}$$

$$S_n = \frac{(1+i)^n - 1}{i}$$

annu. - imm.



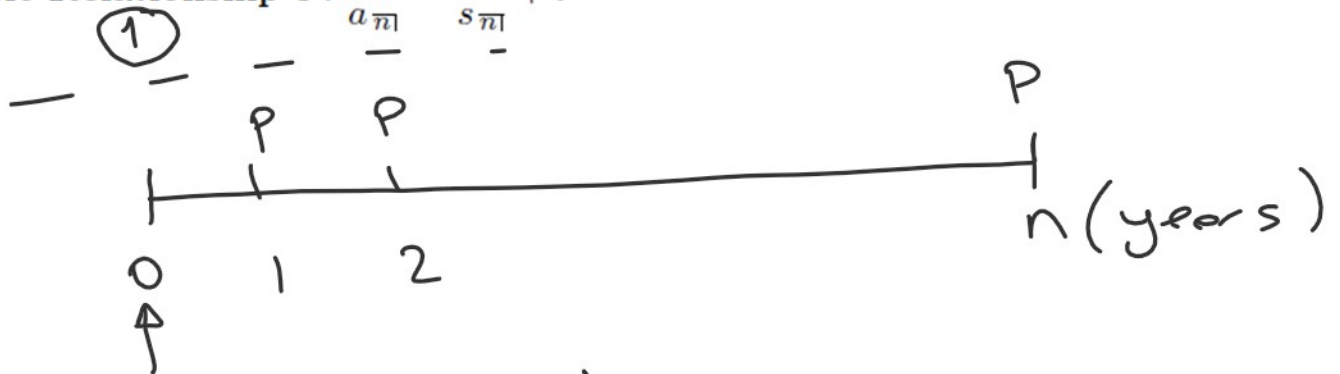


$$s_n \cdot (1+i)^{-n} = a_n$$

$$s_n \cdot v^n = a_n$$

$$a_n \cdot (1+i)^n = s_n$$

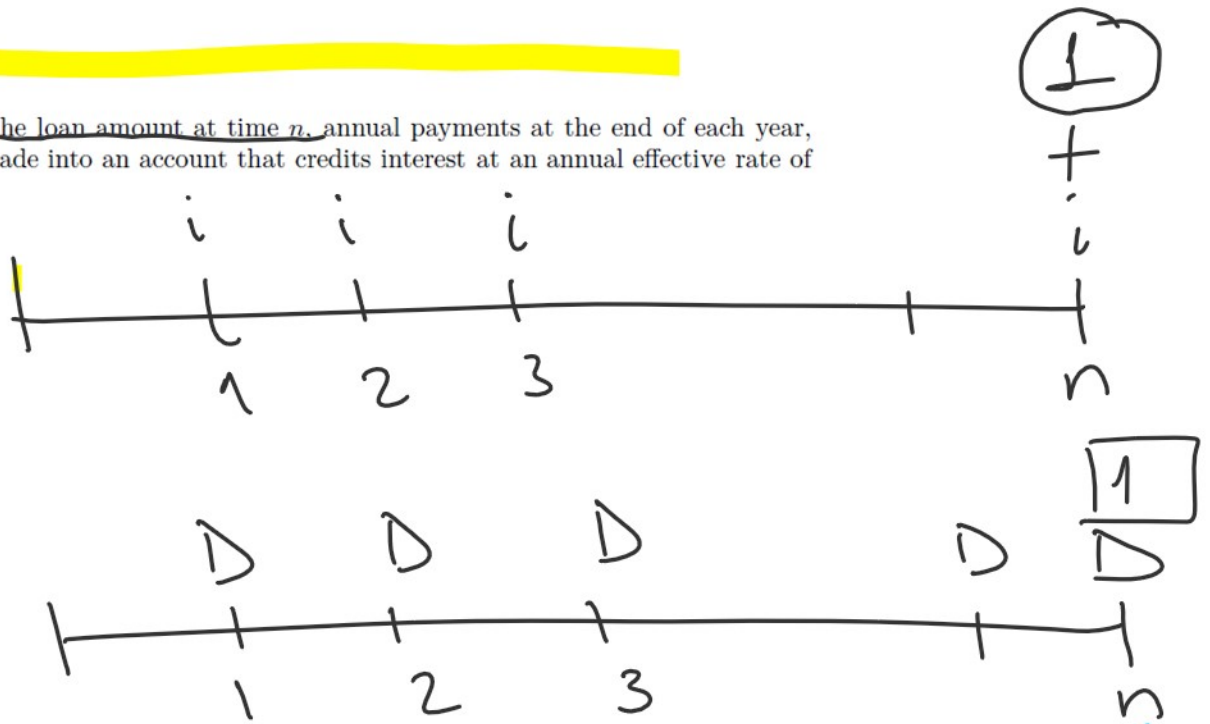
Basic Relationship 3: $\frac{1}{a_n} = \frac{1}{s_n} + i$



$$P \cdot a_n = 1$$

$$P = \frac{1}{a_n}$$

In order to produce the loan amount at time n , annual payments at the end of each year, for n years, will be made into an account that credits interest at an annual effective rate of interest i .



annuity immediate with
the payment $D \cdot (n)$

$$D \cdot s_{\overline{n}|i} = 1$$

$$D = \frac{1}{s_{\overline{n}|i}}$$

$$D + i = \frac{1}{s_{\overline{n}|i}} + i //$$

Note that

$$\begin{aligned} \frac{1}{a_{\overline{n}|i}} &= \frac{i}{1-v^n} \times \frac{(1+i)^n}{(1+i)^n} = \frac{i(1+i)^n}{(1+i)^n - 1} \\ &= \frac{i(1+i)^n + i - i}{(1+i)^n - 1} = \frac{i[(1+i)^n - 1] + i}{(1+i)^n - 1} \\ &= i + \frac{i}{(1+i)^n - 1} = i + \frac{1}{s_{\overline{n}|i}} \end{aligned}$$

Interest Repayment Options

Given a loan of 1, there are 3 options in repaying back the loan over the next n years:

Option 1: Pay back the loan and all interest due at time n .

$$\begin{aligned} \text{Total Interest Paid} &= A(n) - A(0) \\ &= \text{Loan} \times (1+i)^n - \text{Loan} \\ &= \text{Loan} \times [(1+i)^n - 1] \end{aligned}$$

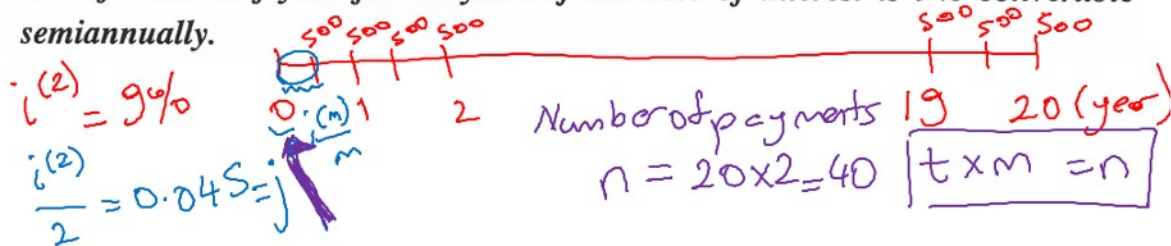
Option 2: Pay at the end of each year, the interest that comes due on the loan and then pay back the loan at time n .

$$\begin{aligned} \text{Annual Interest Payment} &= i \cdot \text{Loan} \\ \text{Total Interest Paid} &= i \cdot \text{Loan} \times n \\ &= \text{Loan} \times (i \cdot n) \end{aligned}$$

Option 3: Pay a level annual amount at the end of each year for the next n years.

$$\text{Annual Payment} = \frac{\text{Loan}}{a_{\overline{n}|i}}$$

Example 3.1 Find the present value of an annuity which pays \$500 at the end of each half-year for 20 years if the rate of interest is 9% convertible semiannually.



The answer is

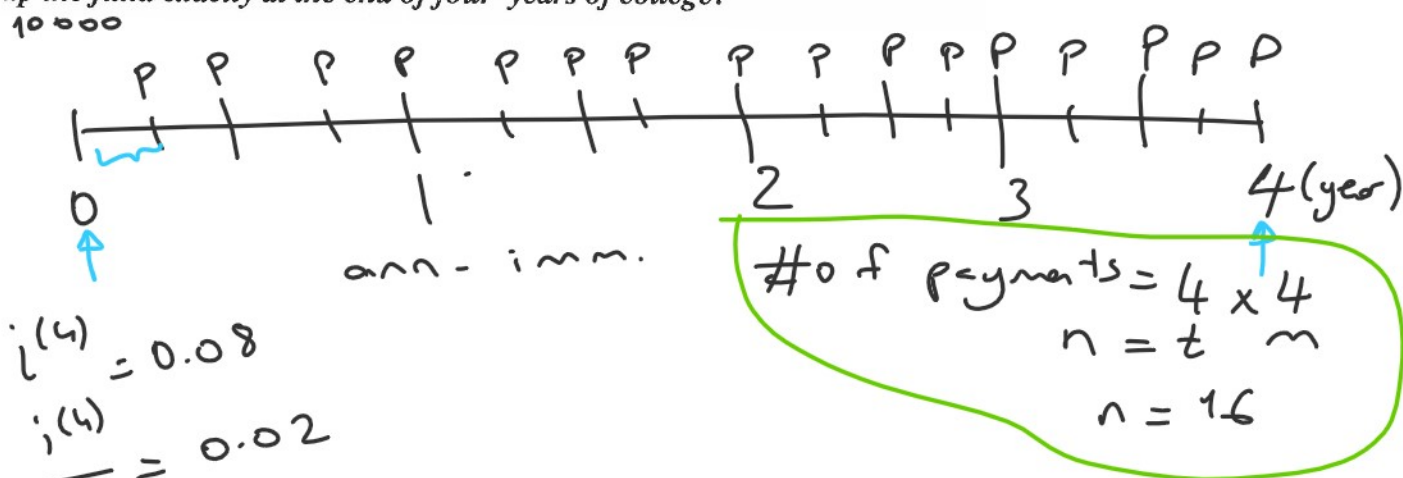
$$500 a_{\overline{40}|0.045}$$

This can be evaluated using formula (3.2) as

$$500 \frac{1 - (1.045)^{-40}}{0.045} = 500(18.40158) = \underline{\underline{\$9,200.79}}$$

$$a_{\overline{n}|} = \frac{1 - (1+i)^{-n}}{i} = \frac{1 - v^n}{i}$$

Example 3.2 If a college freshman invests a \$10,000 gift at 8% per annum convertible quarterly, how much can be withdrawn at the end of every quarter to use up the fund exactly at the end of four years of college?



$$i^{(4)} = 0.08$$

$$\frac{i^{(4)}}{4} = 0.02$$

$$10000 = P \cdot a_{\overline{16}|0.02}$$

$$a_{\overline{n}|} = \frac{1 - v^n}{i}$$

$$a_{\overline{16}|0.02} = \frac{1 - (1.02)^{-16}}{0.02}$$

Let R be the amount of each withdrawal. The equation of value at the date of investment is

$$R a_{\overline{16}|.02} = 10,000.$$

Thus, we have

$$\begin{aligned} R &= \frac{10,000}{a_{\overline{16}|.02}} \\ &= \frac{10,000}{13.57771} \\ &= \$736.50. \end{aligned}$$